

CPSC 8440: Generative AI: Technical and Human Perspectives

Module 1.3.1: Word Embeddings

Word embeddings

- Word embeddings = Vector representations of word meaning

荃者所以在鱼，得鱼而忘荃 Nets are for fish;
Once you get the fish, you can forget the net.
言者所以在意，得意而忘言 Words are for meaning;
Once you get the meaning, you can forget the words
庄子(Zhuangzi), Chapter 26

How do we represent words in NLP?

- n-gram models

$$P(w_1, w_2, \dots, w_n) = \prod_{i=1}^n P(w_i | w_{i-1})$$

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i) + \alpha}{C(w_{i-1}) + \alpha|V|}$$

Each word is just a string
or indices w_i in the
vocabulary list

cat = the 5th word

dog = the 10th word

cats = the 118th word

- Naïve Bayes

$$\hat{P}(w_i | c_j) = \frac{\text{Count}(w_i, c_j) + \alpha}{\sum_{w \in V} \text{Count}(w, c_j) + \alpha|V|}$$

What are some issues with representing words in these ways?

- A feature is just a word identity (= string)
 - Feature 5: ‘The previous word was “terrible”’
 - Requires **exact same word** to be in the training and testing set

“terrible” ≠ “horrible”

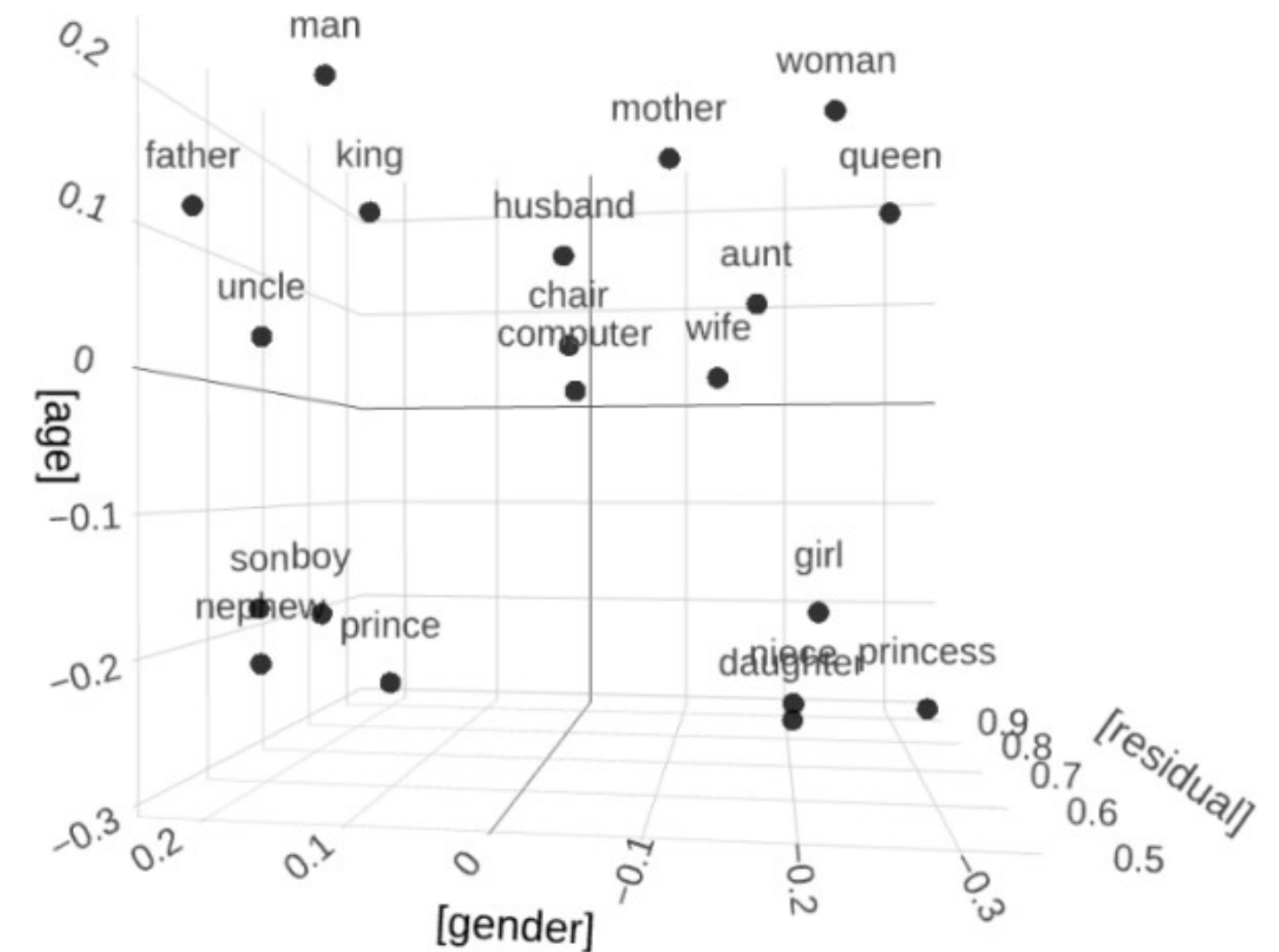
Need for word meaning in NLP models

Word embeddings: word meaning focusing on similarity

Each word = a vector

$$v_{\text{cat}} = \begin{pmatrix} -0.224 \\ 0.130 \\ -0.290 \\ 0.276 \end{pmatrix} \quad v_{\text{dog}} = \begin{pmatrix} -0.124 \\ 0.430 \\ -0.200 \\ 0.329 \end{pmatrix}$$
$$v_{\text{the}} = \begin{pmatrix} 0.234 \\ 0.266 \\ 0.239 \\ -0.199 \end{pmatrix} \quad v_{\text{language}} = \begin{pmatrix} 0.290 \\ -0.441 \\ 0.762 \\ 0.982 \end{pmatrix}$$

Similar words are “**nearby in the vector space**”



(Bandyopaghyay et al.
2022)

Word embeddings

= **Learned** representations from text for representing words

- Input: a large text corpora, V, d
 - V : a pre-defined vocabulary
 - d : dimension of word vectors (e.g. 300)
- Text corpora:
 - Wikipedia + Gigaword 5: 6B tokens
 - Twitter: 27B tokens
 - Common Crawl: 840B tokens
- Output: $f : V \rightarrow \mathbb{R}^d$

$$v_{\text{cat}} = \begin{pmatrix} -0.224 \\ 0.130 \\ -0.290 \\ 0.276 \end{pmatrix} \quad v_{\text{dog}} = \begin{pmatrix} -0.124 \\ 0.430 \\ -0.200 \\ 0.329 \end{pmatrix}$$

$$v_{\text{the}} = \begin{pmatrix} 0.234 \\ 0.266 \\ 0.239 \\ -0.199 \end{pmatrix} \quad v_{\text{language}} = \begin{pmatrix} 0.290 \\ -0.441 \\ 0.762 \\ 0.982 \end{pmatrix}$$

Each word is represented by a low-dimensional (e.g., $d = 300$), real-valued vector

Each coordinate/dimension of the vector doesn't have a particular interpretation

Word embeddings

- Basic property: similar words have similar vectors

word w^* = "sweden"

$$\arg \max_{w \in V} \cos(e(w), e(w^*))$$

Word	Cosine distance
norway	0.760124
denmark	0.715460
finland	0.620022
switzerland	0.588132
belgium	0.585835
netherlands	0.574631
iceland	0.562368
estonia	0.547621
slovenia	0.531408

$\cos(u, v)$ ranges between -1 and 1

Word embeddings

- Basic property: similar words have similar vectors

Nearest words to
frog:

1. frogs
2. toad
3. litoria
4. leptodactylidae
5. rana
6. lizard
7. eleutherodactylus



litoria



leptodactylidae



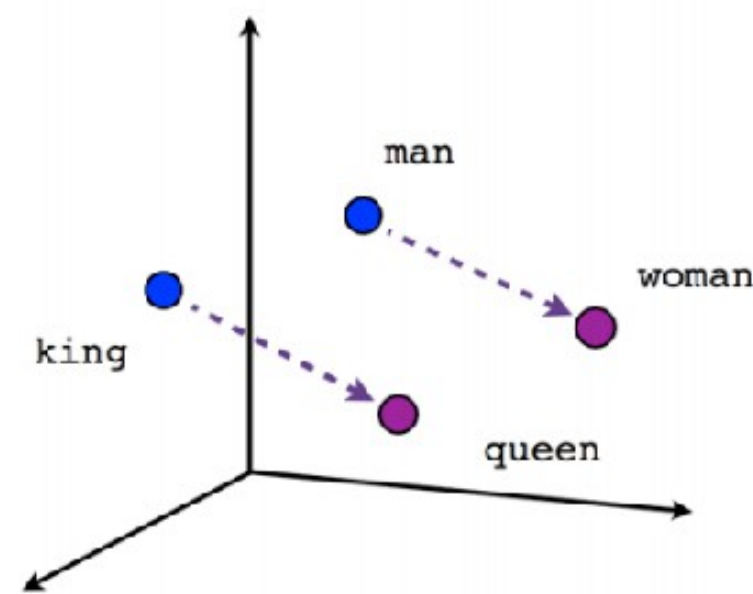
rana



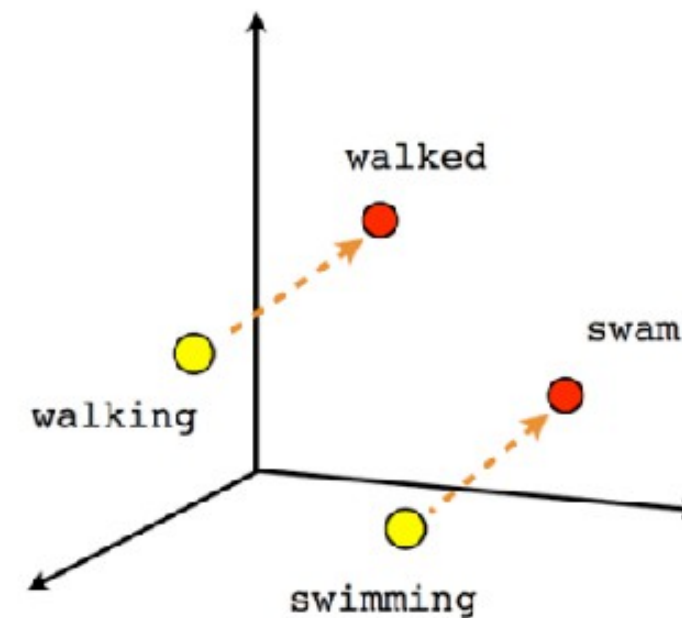
eleutherodactylus

Word embeddings

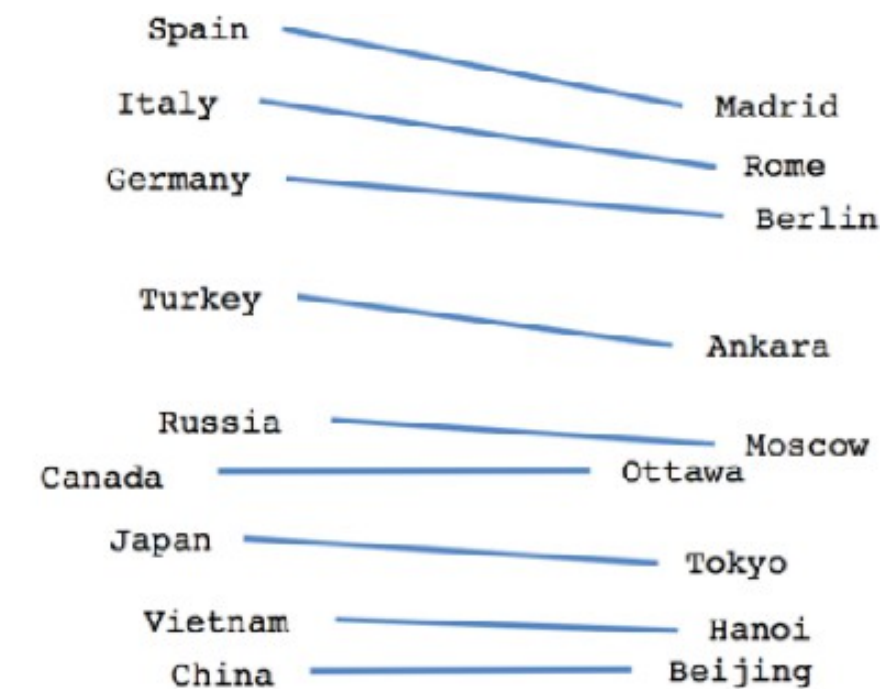
- They have some other nice properties too!



Male-Female



Verb tense



Country-Capital

$$v_{\text{man}} - v_{\text{woman}} \approx v_{\text{king}} - v_{\text{queen}}$$

$$v_{\text{Paris}} - v_{\text{France}} \approx v_{\text{Rome}} - v_{\text{Italy}}$$

Word analogy test: $a : a^* :: b : b^*$

$$b^* = \arg \max_{w \in V} \cos(e(w), e(a^*) - e(a) + e(b))$$

Word embeddings

- They have some other nice properties too!

$$v(\text{cuatro}) \approx Wv(\text{four})$$

